David Nguyen

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Education

University of Michigan **Expected Graduation: May 2027**

B.S.E. in Computer Science | GPA: 3.6/4.0 *Ann Arbor, MI*

Relevant Coursework: Data Structures and Algorithms, Advanced Operating Systems, Computer Architecture, Web Systems, Object Oriented Programming, Logic Design, Discrete Math, Probability, Linear Algebra, Multivariable Calculus

Organizations: Michigan Hackers, Michigan Data Science Team, Institute of Electrical and Electronics Engineers (IEEE)

Experience

Roadsr **October 2025 - Janurary 2026**

Software Developer *Ann Arbor, MI*

- Built and deployed a GitHub Actions CI/CD pipeline for Python services, executing parallelized automated test suites to reduce regression defects and improve deployment reliability
- Implemented automated collision detection tests to validate Time-to-Collision alert thresholds across speed- and distance-dependent edge cases, leveraging Python, Pandas, NumPy, and OpenCV to improve system robustness and real-time reliability

VOID Tech **January 2025 - April 2025**

Software Developer *Ann Arbor, MI*

- Designed and implemented backend infrastructure for a subscription-based streaming platform, enabling users to stream God Said Give 'Em Drum Machines using TypeScript and AWS Amplify for scalability
- Implemented authentication and authorization account management using AWS Cognito for secure sign-ups/logins and tracked user subscriptions status via AWS DynamoDB with PostgreSQL for gated media access
- Integrated a video streaming infrastructure using MUX with signed URLs and thumbnail tokens to address authorized access concerns for paid users

N2K **November 2023 – July 2024**

IT Intern *Fulton, MD*

- Developed an internal website using HubSpot and Adobe Photoshop to advocate 20% of the company to acquire the ISC2 Certified in Cybersecurity certification for higher organizational cybersecurity credibility with IT partners
- Organized client data using Python and Pandas to export client data into readable CSV files and audited company's self-diagnostic products for simpler data analytics to acquire over $200,000 from partnered companies and clients

Projects

Trading Bot | *C++, Python, PyTorch, FastAPI, React.js, TypeScript, PostgreSQL*

- Architected a NASDAQ trading bot with responsive UI featuring charts with adjustable stop-loss values and statistics with execution metrics such as net profit and win rate percentage, backed by PostgreSQL for historical trade data
- Integrated a low-latency C++ backend for trade execution with AI-driven reasoning and NASDAQ market forecasting using the OpenAI API and Hugging Face, enabling detailed strategy analysis and planning

Multiprocessor Thread Library | *C++, Python, Concurrent Programming*

- Implemented a thread library in C++ supporting thread creation, mutexes, and condition variables across multiple CPUs, managing low-level context switching, timer-based and inter-processor interrupts for CPU sleep and wake coordination
- Eliminated race conditions in a concurrent kernel simulation by enforcing atomic guard protocols, validating synchronization correctness, and memory life cycles across multiple CPUs through automated Python stress testing

Instagram Clone | *React.js, TypeScript, Python, Flask, REST APIs, SQL, AWS*

- Developed a RESTful Flask backend implementing authentication, session management, and CRUD APIs for posts, comments, and likes, backed by SQL database integration and deployed on AWS EC2
- Built a dynamic React.js frontend with component-based architecture, client-side routing, and state management to render real-time posts, likes, comments, and user authentication workflows with responsive UI using Tailwind CSS

Habitat AI | *React.js, Next.js, RestAPI, TypeScript, Mastra*

- Developed backend services in Node.js integrating Mastra AI agents and Gemini API to auto-generate an infinitely expanding knowledge graph, dynamically creating related topics from any root node and powering real-time visualization
- Built a Chrome extension to scrape external content and import it as central nodes, enabling users to expand from real-world sources; enhanced exploration by displaying definitions, links, and videos on hover for immediate context

Skills

Languages: C/C++, Python, TypeScript, Java, HTML/CSS/JavaScript, SQL, MATLAB, Assembly, Verilog

Technologies: Node.js, PostgreSQL, PyTorch, Git, Linux, AWS, Azure

Frameworks: React.js, Flask, Next.js, Tailwind CSS, FastAPI, PyTorch, Pandas, NumPy,

Certifications: AWS Cloud Practitioner, Microsoft Azure Fundamentals, ISC2 Certified in Cybersecurity